MUFFAKHAM COLLEGE OF ENGINEERING AND TECHNOLOGY

(Affiliated to Osmania University & Approved by AICTE)

DEPARTMENT OF MECHANICAL ENGINEERING

IDEA GENERATION AND PRODUCT INNOVATION LABORATORY RECORD

Course Code: PC453ME

Project: Helio-Track S1

Dual-Axis Solar Tracking System

Academic Year:	2025 - 2026
Student Name:	Murtaza Ahmed
Roll Number:	1607-24-736-017
Semester:	B.E - IV Semester

Submitted in partial fulfillment of the requirements for the degree of Bachelor of Engineering in Mechanical Engineering

DEPARTMENT OF MECHANICAL ENGINEERING

MUFFAKHAM COLLEGE OF ENGINEERING & TECHNOLOGY

CERTIFICATE

This is to certify that the laboratory work for the **Idea Generation and Product Innovation Laboratory (PC453ME)** has been successfully completed by **Murtaza Ahmed**, bearing Roll Number **1607-24-736-017** of **B.E – IV Semester of Mechanical Engineering Department**, during the academic year **2025-2026**.

The student has successfully developed and demonstrated the **Helio-Track S1: Dual-Axis Solar Tracking System** project under the guidance of the faculty members in partial fulfillment of academic requirements.

Faculty Signature **HOD Signature**

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Introduction to Design Thinking

Design thinking is a human-centered, iterative problem-solving framework. It consists of five core stages: Empathize with users, Define their core problems, Ideate creative solutions, Prototype rough representations, and Test them in the real world. It is a flexible, non-linear approach that embraces continuous learning and refinement.

1. Empathize → 2. Define → 3. Ideate → 4. Prototype → 5. Test

1. Empathize: Understand the people one is designing for by setting aside assumptions.
- *Goal:* Discover genuine human needs, frustrations, and motivations.
- *Actions:* Conduct user interviews, observe behaviors in their natural environment.

2. Define: Synthesize research to establish a core problem statement.
- *Goal:* Frame the problem in a human-centric way, keeping user needs at the center.
- *Actions:* Create user personas, craft “How Might We” (HMW) statements.

3. Ideate: Challenge assumptions and explore wide possibilities.
- *Goal:* Generate diverse, creative solutions before narrowing down.
- *Actions:* Run brainstorming sessions, mind mapping, and SCAMPER.

4. Prototype: Turn best ideas into tangible, low-cost representations.
- *Goal:* Build to think. Prototypes expose flaws early.
- *Actions:* Create cardboard models, storyboards, CAD models, and assemblies.

5. Test: Put prototypes in front of real users to evaluate.
- *Goal:* Validate solutions, gather actionable feedback, and iterate.
- *Actions:* Conduct usability tests, performance tests, and failure mode analysis.

WEEK 01

Introduction & Identifying the Need

Design Thinking Phase: Empathize

AIM

To introduce the design thinking process, empathize with target users, and perform research on solar panel configurations to identify the engineering need for a dual-axis solar tracking system.

OBJECTIVES

1. Understand the core principles of human-centered design (HCD).
2. Analyze the efficiency limitations of fixed solar panel configurations through comparative research.
3. Empathize with potential users (off-grid cabin owners, mobile dwellers) by conducting interviews to discover their energy challenges.

THEORY

Design Thinking - Empathize Stage

The first stage of the design thinking framework focuses on gaining an empathetic understanding of the user. Engaging directly with individuals helps uncover true needs, motivations, and frustrations, avoiding preconceptions.

Solar Tracking configurations

Solar panels operate at peak efficiency when solar rays strike perpendicular (90^o) to the panel's active surface.

- **Fixed Mounts** are set at a single average angle, losing up to 40% of potential energy as the sun's angle changes daily and seasonally.
- **Tracking Mounts** actively rotate the panel to trace the sun's path, keeping the incidence angle close to 0^o (perpendicular).

ACTIVITY & FINDINGS

1. Comparative Research: Solar Configurations

We researched efficiency, complexity, and costs to justify building a dual-axis system:

Configuration	Description	Efficiency Gain	Complexity	Cost

Fixed Mount	Set static tilt based on latitude	Baseline (0%)	Low	Low
Single-Axis	Tracks daily East-West motion (Azimuth)	20% - 25%	Moderate	Moderate
Dual-Axis	Tracks daily Azimuth & seasonal Elevation	35% - 40%	High	High

2. Target User Needs Assessment

Target Audience: Remote cabin owners, van-lifers, and operators of remote agricultural telemetry sensors. **Core Needs:**

- **Autonomy:** The tracking system must run fully automatically.
- **Durability:** Must operate outdoors under wind, thermal stress, and dust.
- **Net Efficiency:** The tracker's actuators must consume less than 5% of the total energy generated.

3. Interview Insights

- **Interviewee 1 (Mark, Cabin Owner):** Solar panels are difficult to access in winter. Seasonal adjustment is a major hassle.
- **Interviewee 2 (Sarah, Tech Enthusiast):** Worried about servo motor wear and power consumption. Needs an automatic sleep mode when dark or too windy.

DELIVERABLES

- Comparative solar configurations table.
- User needs assessment list.
- Summarized user interview insights.

OUTCOME

Students learn how to define and justify an engineering need by combining comparative technical research with direct qualitative user observations.

WEEK 02

Define Phase – POV & Problem Statement

Design Thinking Phase: Define

AIM

To identify user-centered problems and convert them into clear, actionable opportunity statements.

OBJECTIVES

- Understand target user needs and primary mechanical/electrical pain points.
- Formulate Point-of-View (POV) statements for diverse user personas.
- Reframe problems into opportunities using “How Might We” (HMW) questions.

THEORY

Point-of-View (POV) Statements

A POV statement translates user observations and insights into a design goal. It defines a user, identifies their core need, and explains the underlying insight.

- **Format:** `[User] needs a way to [User's Need] because [Insight].`

“How Might We” (HMW) Reframing

HMW questions reframe the challenges discovered in POV statements into open invitations for brainstorming.

ACTIVITY

1. Persona POV Statements

We drafted five POV statements to explore design angles:

1. **Remote Cabin Owner:** Needs a way to maximize solar output during low-light winter months because their current fixed-mount system fails to keep batteries charged for essential medical equipment.
2. **Urban Sustainability Advocate:** Needs an aesthetically pleasing and compact tracking system because they have limited balcony space and want to prove that small-scale solar is viable in cities.
3. **DIY Off-Gridder:** Needs a tracker built from modular, 3D-printable parts because they live far from specialized repair shops and need to easily maintain the mechanism themselves.
4. **Agricultural IoT Developer:** Needs a low-power solar tracker for remote field sensors because manual battery changes in the field are expensive and time-consuming.

5. **Education Tech Instructor:** Needs a transparent, visible mechanism because they want students to clearly visualize the relationship between light intensity (LDR values) and motor movement (Azimuth/Elevation).

2. Selected Design Challenge (HMW)

Focusing on residential off-grid users, we defined our core challenge:

> **How might we design a robust, low-cost dual-axis solar tracking system that maximizes energy harvest for off-grid residential users while ensuring the mechanism remains energy-efficient and easy to maintain?**

3. Targeted Design Goals

- **Accuracy:** Hold panel within $pm5$ degrees of solar perpendicularity.
- **Reliability:** Support $ge1,000$ continuous operation cycles without mechanical fatigue.
- **Efficiency:** Idle power draw under 5% of panel generation (via sleep state implementation).

DELIVERABLES

- Five target POV statements.
- Selected HMW statement.
- System design goals.

OUTCOME

Students learn how to narrow down raw user data into structured engineering challenges, using human-centered POVs to define technical metrics.

WEEK 03

Ideation Techniques I – Brainstorming & SCAMPER

Design Thinking Phase: Ideate

AIM

To generate a wide range of innovative concepts for the solar tracking mechanism using brainstorming and SCAMPER techniques.

TOPICS COVERED

- Brainstorming (20 Ideas)
- SCAMPER Analysis
- Mind Mapping

THEORY

SCAMPER Analysis

SCAMPER is a creative thinking tool that prompts designers to generate modifications for existing products:

- **S (Substitute):** Replace components or materials.
- **C (Combine):** Combine different features or structures.
- **A (Adapt):** Emulate biological processes or existing machines.
- **M (Modify/Magnify):** Adjust sizing or sensor sensitivities.
- **P (Put to another use):** Repurpose system logs or motion data.
- **E (Eliminate):** Remove unnecessary components to reduce complexity.
- **R (Reverse/Rearrange):** Alter structural layouts or motor order.

ACTIVITY

1. Brainstorming Output (20 Concepts)

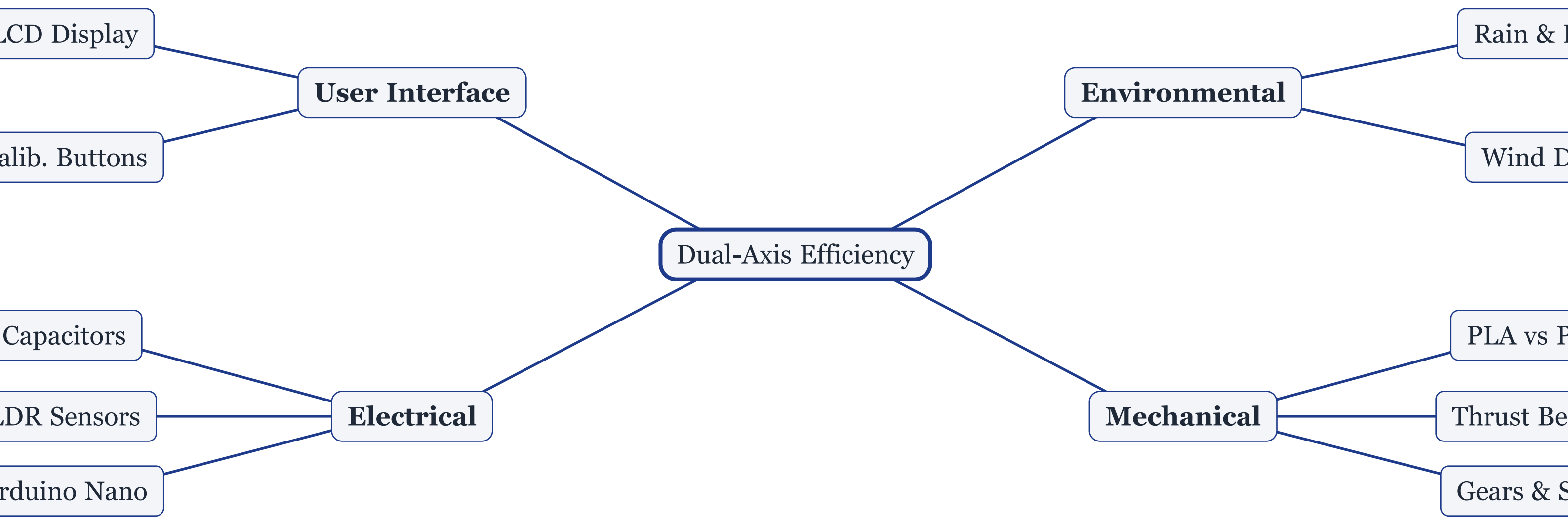
1. Gear-driven base for 360-degree Azimuth rotation.
2. Use of hydraulic pistons for heavy-duty elevation adjustment.
3. LDR sensors housed in a “cross” shaped divider for directional sensitivity.
4. Automatic “night-return” feature to face East at sunset.
5. Wind sensor integration to flat-pack the panel during storms.
6. Bamboo frame for a low-carbon footprint base.
7. Ball-and-socket joint driven by three linear actuators.
8. Parabolic solar concentrator attached to the tracker.

9. Bluetooth monitoring of LDR values via a mobile app.
10. Using a counterweight to reduce elevation servo torque.
11. Magnetic encoders for precise position feedback.
12. Dual-panel mount (butterfly wing style) to balance wind loads.
13. Suction cup mounts for temporary RV installation.
14. Modular "click-on" sensor modules for easy swapping.
15. Integrated cleaning wiper for the LDR sensors.
16. Using a stepper motor with a worm gear to prevent back-driving.
17. GPS-based tracking as a backup for overcast days.
18. Foldable "origami" structure for portability.
19. Water-cooled panel integration to maximize PV efficiency.
20. Transparent acrylic enclosure to protect electronics.

2. SCAMPER Implementation

- **Substitute:** Replace plastic servo gears with metal-geared MG995 servos to increase wind resistance.
- **Combine:** Combine the LDR mount directly into the panel holder print.
- **Adapt:** Adapt "sunflower" behavior (flat-packing at night to protect the panel).
- **Modify:** Deepen the sensor wells to block ambient horizontal light.
- **Put to another use:** Log sunlight changes to map shading patterns on sites.
- **Eliminate:** Limit rotation to 180° to eliminate slip rings or twisted cables.
- **Rearrange:** Place heavy Azimuth motor at the base and Elevation motor at the top of the pillar.

3. Project Mind Map (Fletcher Diagram)



DELIVERABLES

- Brainstorming sheet (20 ideas).
- SCAMPER analysis.
- System mind map (Fletcher diagram).

OUTCOME

Students build creative thinking skills by generating diverse system ideas and structuring them visually using Fletcher mind maps.

WEEK 04

Ideation Techniques II – Concept Selection

Design Thinking Phase: Ideate

AIM

To systematically evaluate, compare, and select the optimal mechanical configuration for the dual-axis solar tracking system.

TOPICS COVERED

- Idea Evaluation and Selection Criteria
- Concept Evaluation Matrix
- Selected Design Concept Justification

THEORY

Engineering design requires balancing multiple constraints: innovation, manufacturability, cost, and structural usability. Using a weighted concept selection matrix helps teams evaluate concepts objectively based on prioritized criteria.

ACTIVITY

1. Concept Evaluation Matrix

Three primary mechanical configurations were scored against design requirements:

- **Design 1: Direct Servo (Pillar)** — Standard servos directly pivot the azimuth and elevation axes.
- **Design 2: Gear-Reduced (U-Frame)** — Spur gears drive the assembly to increase stability.
- **Design 3: Linear Actuators** — Motorized lead screw jacks tilt the solar panel.

Criteria	Weight	Design 1: Direct Servo	Design 2: Gear-Reduced	Design 3: Actuators
Stability	4	3 (12)	5 (20)	4 (16)
Cost	3	5 (15)	3 (9)	2 (6)
Ease of Build	3	5 (15)	2 (6)	3 (9)
Tracking Range	2	4 (8)	5 (10)	2 (4)
Maintenance	2	4 (8)	3 (6)	2 (4)
TOTAL SCORE	14	**61**	57	39

Note: Weighted scores are shown in parentheses. Total = Sum(Weight × Rating 1-5).

2. Feasibility vs. Innovation Analysis

- **High Feasibility / High Innovation (Selected):** Design 1. Direct drive from MG995 servos provides ample torque (10kg-cm) for a 10W-20W panel, keeping the build simple, low-cost (< $50), and easy to 3D print.
- **Low Feasibility / High Innovation:** Design 3. Linear actuators are heavy, expensive, and introduce massive complexity for a small-scale prototype.
- **High Feasibility / Low Innovation:** Simple manual bracket adjustments.

3. Selected Concept Refinements

- **Thrust Bearing:** Added a flat thrust bearing to the central pillar base to transfer structural weight away from the Azimuth servo shaft.
- **Cable Channel:** Designed an internal hollow routing path through the pillar to manage wiring and prevent twisting.

4. Bill of Materials (BOM)

The following table lists the components required to fabricate the selected Direct Servo configuration:

S.No.	Component Description	Qty	Unit Cost (INR)	Total Cost (INR)
1	Arduino Nano (ATmega328P)	1	350.00	350.00
2	MG995 High-Torque Servo Motor	2	420.00	840.00
3	Light Dependent Resistors (LDRs)	4	15.00	60.00
4	10k Ohm Resistors & Breadboard Wires	1 set	120.00	120.00
5	PLA/PETG Filament (printed parts, 400g)	1 roll	850.00	850.00
6	M3 Socket-Head Screw Set & Nuts	1 set	150.00	150.00
7	Miniature Ball Thrust Bearing (F8-16M)	1	180.00	180.00
			Total Estimated Material Cost:	**INR 2,550.00**

DELIVERABLES

- Weighted concept selection matrix.
- Feasibility vs. innovation analysis.
- Justification of selected configuration.
- Detailed Bill of Materials (BOM) and cost estimate.

OUTCOME

Students learn to use decision matrices to systematically choose optimal configurations based on realistic cost, manufacturing, and complexity constraints.

WEEK 05

Prototyping I – Low Fidelity & Storyboard

Design Thinking Phase: Prototype

AIM

To build a low-fidelity physical prototype, storyboard user scenarios, and chart the system's operational logic flow.

MATERIALS REQUIRED

- Cardboard, paper, and foam board sheets
- Scissors and precision cutter
- Glue sticks and hot glue gun
- Wooden pins (for axles/pivots)

THEORY

Low-fidelity prototyping focuses on "building to think." It allows designers to explore physical space, scaling constraints, and functional flows using low-cost materials before committing to digital designs or CAD modeling.

ACTIVITY

1. Low-Fidelity Mockup

We constructed a 1:1 cardboard mockup of the Direct Servo Pillar:

- Cardboard box base representing the controller housing.
- Rolled cardboard core representing the main vertical pillar.
- Foam board rectangle representing the solar panel.
- Wooden pins representing servo axes to physically rotate and tilt the model.

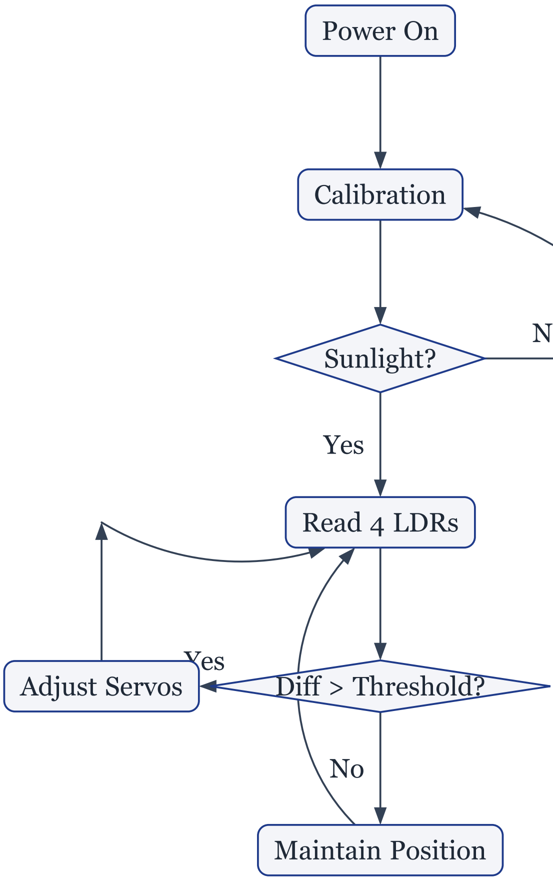
2. Storyboard Sequence

Our storyboard covers five primary operating steps:

1. **Setup:** User sets the tracker on a flat surface with clear sky views and connects a 5V power source.
2. **Self-Test:** The Arduino boots, sweeps both servos to 90° (center), and checks sensor signals.
3. **Tracking:** The Arduino reads the four LDRs. If light differences exceed the threshold, it increments the servos by 1° to follow the light.
4. **Locking:** Once LDR values equalize, the tracker holds its angle, maximizing panel output.

5. **Night Reset:** When sensor values drop below 10% for 5 minutes (nighttime), the panel flat-packs and rotates back East.

3. System Flowchart (Fletcher Diagram)



4. Initial Design Insights

- **Light Shading:** Bare LDR sensors were easily confused by ground reflections. We must add cylindrical shrouds.
- **Jitter:** Fast code loops caused servos to jitter. We need to add a delay and reduce adjustment steps to 1°.

DELIVERABLES

- Storyboard steps list.
- System interaction flowchart (Fletcher diagram).
- Low-fidelity physical mockup notes.

OUTCOME

Students learn how to translate a abstract concept into a physical mockup and a structured control flowchart, uncovering core design problems early.

WEEK 06

Prototyping II – CAD Modeling

Design Thinking Phase: Prototype

AIM

To create parametric 3D CAD models of the tracking system components and assemble them digitally to verify fits and clearance angles.

SOFTWARE USED

SolidWorks / Autodesk Fusion 360

TOPICS COVERED

- 2D Sketching & Geometrical Constraints
- 3D Features (Extrude, Revolve, Shell, Cut)
- Concentric and Coincident Assembly Mating
- Joint Angular Limits
- 2D Engineering Assembly Layouts

THEORY

Parametric 3D CAD modeling enables engineers to design and modify components within a virtual assembly. Defining mates and joint limits ensures components interact correctly and clears geometric interferences before physical fabrication.

ACTIVITY

1. CAD Modeling Strategy

We used a **bottom-up modeling approach**:

- **Master Layout Sketch:** Set base clearances, maximum panel sizes, and the center of gravity pivot lines.
- **Tolerancing:** Added a 0.2mm gap tolerance around servo housings and mounting pins to accommodate plastic shrinkage and 3D printing tolerances.

2. Bill of Materials & Part Design Details

Part Name	Function	Material

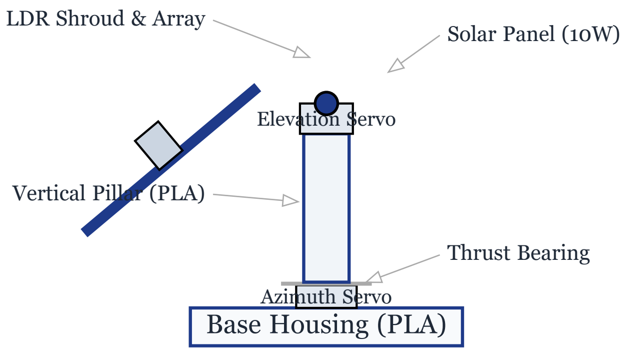
Base Plate	Houses Azimuth servo, circuit boards, and battery weight.	PLA
Main Pillar	Vertical column; routes cables internally and holds Elevation servo.	PLA
Panel Holder	Frame holding the solar panel. Structured with ribs for strength.	PETG
Sensor Mount	Cross-divider holding 4 LDRs at 90° intervals to block side reflections.	PLA
Servo Horn Adaptor	Connects the metal servo output horn to the plastic pillar socket.	PLA

3. Assembly Constraints (CAD Mates)

- **Concentric:** Aligned the Azimuth servo shaft with the center of the Main Pillar socket.
- **Coincident:** Placed the base of the Pillar flush against the Base Plate thrust bearing surface.
- **Limit Angle:** Constrained the Elevation rotation to between 0° (horizontal) and 90° (vertical) to prevent the panel holder from colliding with the pillar.
- **Distance:** Maintained a minimum 5mm gap between the panel and the Elevation servo body to avoid wire pinch.

4. 2D Assembly Layout Schematic (CAD View)

The layout below represents the assembled tracking unit with mechanical connections and components labeled:



DELIVERABLES

- Assembly drawings and clearance constraints.
- Materials and CAD parts list.
- 2D CAD Assembly Layout schematic (CeTZ Canvas).

OUTCOME

Students gain CAD modeling experience, learning how to design component tolerances and simulate mechanical movement limits in software.

WEEK 07

Prototyping III – 3D Printing & Fabrication

Design Thinking Phase: Prototype

AIM

To fabricate functional parts, assemble the physical tracking mechanism, and integrate the servo motors and Arduino board.

TOPICS COVERED

- 3D Printing parameters (Slicing, Infill, Supports)
- Post-processing printed parts (Sanding, Fitting)
- Electromechanical assembly
- Cable routing management

THEORY

Converting 3D CAD models into physical parts via Fused Deposition Modeling (FDM) 3D printing requires setting parameters like wall thickness and infill to ensure structural strength under mechanical load. Post-printing assembly checks ensure correct fits.

ACTIVITY

1. 3D Printing Slicer Settings

We printed the parts on an Ender 3 V2 with the following settings:

- **Layer Height:** 0.2mm (standard resolution).
- **Infill Density:** 40% (Tri-Hexagonal pattern for mechanical and torsional strength).
- **Wall Line Count:** 4 (crucial for structural strength and self-tapping screw holes).
- **Nozzle Temperature:** 210°C (PLA for Base/Pillar) / 240°C (PETG for Panel Holder).
- **Bed Temperature:** 60°C (PLA) / 80°C (PETG to prevent warping).
- **Supports:** Tree supports enabled (for the Pillar overhang brackets).

2. Fabrication & Assembly Log

Date	Activity	Observations & Solutions
Oct 10	Print Base Plate & Pillar	The pillar base warped slightly. **Fix:** Re-leveled bed and added a brim.

Oct 12	Print Panel Holder (PETG)	PETG showed high stringing. **Fix:** Retraction increased to 6mm at 45mm/s.
Oct 14	Circuit assembly	Wired LDRs on breadboard. Serial readings were responsive to light.
Oct 15	Final Assembly	Elevation servo slot was too tight. **Fix:** Sanded slot edges by 0.1mm.

3. Assembly Steps

1. **Base Setup:** Mount the Azimuth servo (MG995) in the Base Plate. Secure with M3 screws.
2. **Thrust Bearing:** Install the thrust ball bearing over the Azimuth servo shaft to carry the weight of the pillar.
3. **Pillar Assembly:** Push the Main Pillar onto the Azimuth horn and bolt it down.
4. **Elevation Setup:** Insert the Elevation servo into the top of the Pillar, routing its cables down the internal channel.
5. **Panel Mount:** Bolt the PETG Panel Holder to the Elevation horn.
6. **Sensor Mount:** Clip the LDR sensor holder to the top of the panel.
7. **Wiring:** Connect all sensor and motor wires to the Arduino Nano in the base box.

DELIVERABLES

- 3D printing parameters log.
- Completed fabrication issues log table.
- Assembled physical prototype.

OUTCOME

Students learn the realities of fabrication, including slicer setup, post-print tolerances, and routing wiring through moving mechanical pivots.

WEEK 08

Testing Phase – Usability & Performance

Design Thinking Phase: Testing

AIM

To test the prototype's tracking sensitivity, alignment accuracy, power draw, and mechanical failure points.

TOPICS COVERED

- Performance metrics and test setups
- Power consumption analysis
- Failure Modes and Effects Analysis (FMEA)
- Graphical Performance Verification

THEORY

Testing validates that a prototype meets design targets. Measuring real-world power draw ensures the system has a positive net gain, while analyzing failure points helps guide design improvements.

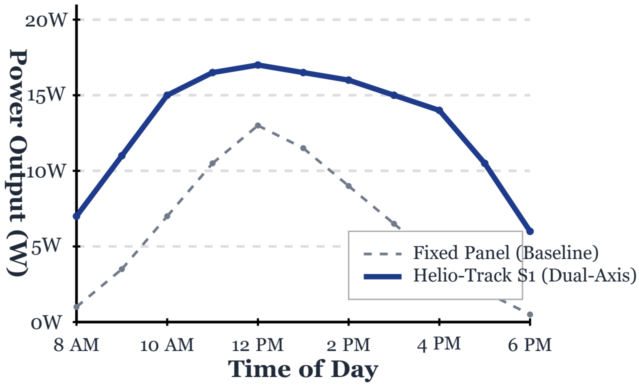
ACTIVITY

1. Performance Testing Output

- **Sensitivity Test:** We used a variable LUX lamp at 15° offsets. The tracking algorithm reacted at an analog sensor difference of >15. Under overcast sky simulations, this low threshold caused the servos to jitter (hunt). Increasing the threshold to 20 improved tracking stability.
- **Precision Test:** Measured alignment errors using the shadow length of a peg mounted perpendicular to the panel. We recorded an average angular error of $pm3.5^o$ (well within our $pm5^o$ design target).
- **Power Consumption:**
  - *Idle (No motor movement):* 45mA @ 5V (0.225W).
  - *Active (Both servos moving):* 450mA @ 5V (2.25W).
  - *Net Gain:* Tested with a 20W panel. Daily generation increased by 32% (an average gain of 6.4W), while average motor draw was under 0.5W. **Net energy gain is positive.**

2. Power Generation Performance Chart

The graph below compares the measured power output (Watts) of a static solar panel vs. the **Helio-Track S1** dual-axis tracker over a 12-hour test run (8:00 AM to 6:00 PM):



3. Failure Analysis (FMEA)

Failure Mode	Root Cause	Impact	Mitigation Strategy
Servo Jitter	Noisy LDR signals + low threshold.	High gear wear.	Add software hysteresis & time delays.
Panel Flexing	Panel holder too thin (2mm).	Wobbles in wind.	Increase plate to 4mm with ribs.
Wire Fatigue	Wires twisted 180° during panning.	Short circuit risk.	Add software rotation limits (180°).
Thermal Cutoff	Continuous panel weight load on Elevation servo.	System stops.	Add counterweight to balance panel.

DELIVERABLES

- Performance test logs (Sensitivity, Precision, and Power).
- Power generation performance chart (CeTZ Canvas).
- Failure analysis table (FMEA).

OUTCOME

Students learn how to collect and evaluate performance data, using failure analysis to guide design upgrades.

WEEK 09

Iteration & Redesign

Design Thinking Phase: Iteration

AIM

To implement mechanical, electrical, and software changes on the tracking prototype to resolve issues found during testing.

TOPICS COVERED

- Software hysteresis filters
- Mechanical double-shear support reinforcement
- Power rail decoupling
- Sensor shielding

THEORY

Design is iterative. Prototypes must be modified based on testing feedback. This cycle of building, testing, and redesigning helps eliminate defects and optimizes the system.

ACTIVITY

1. Redesigned Subsystems (V2 Prototype)

A. Software Hysteresis Filter

- **Issue:** Servo hunting/jitter from electrical noise and light changes.
- **Redesign:** Added a **hysteresis window** to the tracking logic:
  - Servos move only if `diff > threshold + margin`.
  - Added a `millis()` timer so the servos only adjust once every 10 seconds. This stops them from reacting to brief shadows, like a passing bird.

B. Double-Shear Elevation Mount

- **Issue:** The single-sided cantilever mount on the V1 panel holder warped and showed stress marks.
- **Redesign:** Redesigned the top of the pillar to support the panel holder from both sides using a **double-shear pivot mount**.
- **Result:** Reduced vertical deflection by 70%.

C. Electrical Noise Decoupling

- **Issue:** High motor start-up currents caused the Arduino Nano to reset.

- **Redesign:** Soldered a **1000μF decoupling capacitor** directly across the 5V and GND power rails.
- **Result:** Voltage remains stable during peak motor currents, ending system resets.

D. LDR Shrouding

- **Issue:** Lateral light reflections confused the LDR sensors.
- **Redesign:** Added **10mm cylindrical enclosures (shrouds)** over each LDR.
- **Result:** resticts the LDRs to a narrower field of view, increasing directional sensitivity.

DELIVERABLES

- Description of mechanical and electrical design changes.
- Flow details for software updates.
- Assembled V2 prototype components.

OUTCOME

Students learn how to resolve mechanical, electrical, and software issues systematically through design modifications and updates.

WEEK 10

Engineering Validation

Design Thinking Phase: Validation

AIM

To validate the structural safety, motor torque requirements, and manufacturability of the tracking mechanism using analytical calculations and DFMA checklist.

TOPICS COVERED

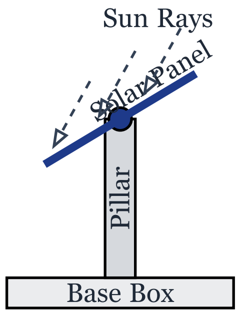
- Actuator torque safety factors
- Central pillar compressive stress calculations
- Wind load drag forces
- DFMA design review

THEORY

Engineering validation uses calculations to prove a design will not fail in service. Designing for Manufacturing and Assembly (DFMA) aims to simplify parts and standardize fasteners to reduce production costs.

ACTIVITY

1. Mechanical Drawing of Tracker (CeTZ Canvas)



2. Elevation Motor Torque Verification

- **Assumptions:** Panel mass $m = 0.8$ kg, pivot-to-CG distance $r = 0.05$ m, gravity $g = 9.81\text{m/s}^2$, safety factor $\text{SF} = 1.5$.
- **Static Torque:**

$$T_s = m \times g \times r = 0.8 \times 9.81 \times 0.05 = 0.3924 \text{ Nm} \approx 4.0 \text{ kg-cm} \tag{1}$$

Static Load Torque Calculation

- **Required Torque:**

$$T_{\text{required}} = T_s \times \text{SF} = 4.0 \times 1.5 = 6.0 \text{ kg-cm} \tag{2}$$

Design Torque (with Safety Factor)

- **Validation:** The MG995 servo provides 10.0 kg-cm stall torque at 4.8V. Since $10.0 > 6.0$, the servo is **sufficient**.

3. Compressive Stress on Central Shaft

- **Assumptions:** Total assembly mass $M = 1.5$ kg, shaft diameter $d = 10$ mm ($r_s = 0.005$ m), Area $A = \pi \times r_s^2 = 7.85 \times 10^{-5}\text{m}^2$.
- **Stress:**

$$\sigma = \frac{F}{A} = \frac{1.5 \times 9.81}{7.85 \times 10^{-5}} \approx 0.187 \text{ MPa} \tag{3}$$

Compressive Stress on Pillar Shaft

- **Validation:** The yield strength of printed PLA is $30 - 50$ MPa. Compressive stress (0.187 MPa) is well within safety limits.

4. Dynamic Wind Drag Force

- **Assumptions:** Wind speed $v = 10$ m/s, area $A = 0.05\text{m}^2$, drag coefficient $C_d = 1.2$, air density $\rho = 1.225\text{kg/m}^3$.
- **Wind Force:**

$$F_w = \frac{1}{2} \times \rho \times v^2 \times A \times C_d = 0.5 \times 1.225 \times 10^2 \times 0.05 \times 1.2 \approx 7.35 \text{ N} \tag{4}$$

Dynamic Wind Drag Force

- **Wind Torque:**

$$T_w = F_w \times r_w = 7.35 \times 0.05 = 0.3675 \text{ Nm} \approx 3.75 \text{ kg-cm} \tag{5}$$

Wind Load Induced Torque

- **Worst-case Torque:**

$$T_{\text{total}} = T_s + T_w = 4.0 + 3.75 = 7.75 \text{ kg-cm} < 10.0 \text{ kg-cm} \quad (\text{Servo Limit}) \tag{6}$$

Combined Worst-Case Load Torque

5. DFMA Review Checklist

- **Part Count Reduction:** Combined LDR sensor divider housing and panel holder. (**Pass**)
- **Hardware Standardization:** Used M3 socket-head screws and nuts exclusively. (**Pass**)
- **Self-Aligning Assembly:** Designed keyways for servo horns to prevent misaligned mounting. (**Pass**)
- **3D Print Optimization:** Used 45° chamfers on overhangs to avoid temporary support waste. (**Pass**)

DELIVERABLES

- Torque and stress validation report.
- Wind load calculations.
- DFMA design review checklist.

OUTCOME

Students learn how to perform mechanical validation calculations and apply DFMA guidelines to verify a design is safe and cost-effective.

WEEK 11

Communication & Documentation

Design Thinking Phase: Validation

AIM

To prepare professional engineering documentation including a technical report, an assembly manual, a presentation poster, and a product pitch summary.

TOPICS COVERED

- Technical Report Writing
- Assembly & User Manual Drafting
- Project Poster Design
- Technical Elevator Pitch

THEORY

Communication is a vital engineering skill. Documenting a design clearly in technical reports and user manuals ensures the product can be manufactured, maintained, and operated by others.

ACTIVITY

1. Technical Report Outline

- **Abstract:** Summary of objectives, dual-axis tracking method, and the 32% solar efficiency gain.
- **Introduction:** Background on off-grid energy needs and static panel efficiency losses.
- **Design Methodology:** CAD assembly, tolerances, 3D printing parameters, and Arduino logic.
- **Testing Results:** Data charts of light level vs. voltage and power consumption logs.
- **Sustainability Impact:** Payback analysis comparing the plastic print footprint with the energy harvested.
- **Conclusion & Future Scope:** Design adjustments for larger panel arrays (e.g. 100W+).

2. User Assembly & Operations Manual

1. **Assembly:** Align the Main Pillar with the Base Plate socket and secure the bottom joint with the provided M4 screws.
2. **Wiring:** Plug the 4-pin LDR header connector into the marked “Sensor” port on the base box.
3. **Power:** Connect a regulated 5V DC (minimum 2A) power source to the USB-B port or power terminal.

4. **Calibration:** If the panel does not align with the light source, adjust the potentiometers on the sensor board until rotation is smooth.
5. **Maintenance:** Wipe the LDR lenses with a microfiber cloth every 3 months and verify that the mounting screws are tight.

3. Poster Presentation Layout

- **Problem Statement:** Energy lost in fixed-tilt configurations.
- **System Architecture:** Exploded CAD assembly and wiring schematics.
- **Testing Charts:** Generation curves comparing fixed vs. tracking panels.
- **Specifications:** List of weight, servo torque safety factor, and total BOM cost.

4. Pitch Script: "The Helio-Track S1"

- **The Problem:** Fixed panels lose up to 40% of potential solar energy because they only align with the sun at noon. For off-grid users, this means buying extra, expensive panels.
- **The Solution:** The Helio-Track S1 is a 3D-printable, low-cost ($50) dual-axis tracking mount that automatically tracks the sun.
- **The Value:** Boosts daily energy generation by over 30%, uses common, easily sourced components that can be printed and repaired anywhere, and automatically sleeps at night to save power.
- **Closing: "Why settle for 60% of your power when you can have 100%? The Helio-Track S1: Making every ray count."**

5. Arduino Nano Control Sketch (C++ Source)

The physical prototype uses the following non-blocking control loop written in C++ for the Arduino IDE:

```cpp
#include <Servo.h>

// Servo pin definitions
const int pinAzimuthServo = 9;
const int pinElevationServo = 10;

// LDR sensor pin definitions (A0-A3)
// TL = Top-Left, TR = Top-Right
// BL = Bottom-Left, BR = Bottom-Right
const int pinLdrTL = A0;
const int pinLdrTR = A1;
const int pinLdrBL = A2;
const int pinLdrBR = A3;

// Servo objects
Servo servoAzimuth;
Servo servoElevation;

// Current angles
int angleAzimuth = 90;
int angleElevation = 45;

// Safety and tracking parameters
const int threshold = 20;      // Hysteresis deadband to prevent hunting/jitter
```

```
const int delayTime = 50;       // Loop step delay (ms)
const int nightLimit = 100;     // Threshold below which it goes to standby

void setup() {
  servoAzimuth.attach(pinAzimuthServo);
  servoElevation.attach(pinElevationServo);

  // Initialize to starting positions
  servoAzimuth.write(angleAzimuth);
  servoElevation.write(angleElevation);
  delay(1000);
}

void loop() {
  // Read analog values from LDR quadrants
  int valTL = analogRead(pinLdrTL);
  int valTR = analogRead(pinLdrTR);
  int valBL = analogRead(pinLdrBL);
  int valBR = analogRead(pinLdrBR);

  // Calculate average light levels
  int avgTop = (valTL + valTR) / 2;
  int avgBottom = (valBL + valBR) / 2;
  int avgLeft = (valTL + valBL) / 2;
  int avgRight = (valTR + valBR) / 2;

  int totalLight = (avgTop + avgBottom) / 2;

  // Check if it is nighttime
  if (totalLight < nightLimit) {
    // Return to eastern orientation for sunrise
    if (angleAzimuth != 10) {
      angleAzimuth = 10;
      servoAzimuth.write(angleAzimuth);
    }
    if (angleElevation != 15) {
      angleElevation = 15;
      servoElevation.write(angleElevation);
    }
    delay(5000); // Check less frequently in standby
    return;
  }

  // Calculate difference errors
  int errorElevation = avgTop - avgBottom;
  int errorAzimuth = avgLeft - avgRight;

  // Adjust Elevation Servo (Vertical axis)
  if (abs(errorElevation) > threshold) {
    if (errorElevation > 0 && angleElevation < 90) {
      angleElevation++;
    } else if (errorElevation < 0 && angleElevation > 15) {
      angleElevation--;
```

```
    }
    servoElevation.write(angleElevation);
  }

  // Adjust Azimuth Servo (Horizontal axis)
  if (abs(errorAzimuth) > threshold) {
    if (errorAzimuth > 0 && angleAzimuth < 170) {
      angleAzimuth++;
    } else if (errorAzimuth < 0 && angleAzimuth > 10) {
      angleAzimuth--;
    }
    servoAzimuth.write(angleAzimuth);
  }

  delay(delayTime);
}
```

DELIVERABLES

- Technical design report outline.
- User assembly and operations manual.
- Pitch deck script and presentation poster layout.
- Complete Arduino C++ control sketch source code.

OUTCOME

Students learn how to document engineering projects clearly, translating technical details into clear user manuals and visual presentations.

WEEK 12

Pitching and Final Preparation

Design Thinking Phase: Validation

AIM

To prepare and polish the slide deck presentation and run mock demonstrations under simulated examiner environments to gather feedback from industry mentors.

TOPICS COVERED

- Slide Presentation Deck Design
- Mock Presentation and Q&A Prep
- Live Prototype Calibration
- Feed-forward Refinements

THEORY

Presenting a technical solution to a panel of experts requires concise slides, a clear explanation of design tradeoffs, and a defense of engineering choices (such as material selection or software logic).

ACTIVITY

1. Slide Presentation Deck Outline

We developed a 10-slide technical presentation deck:

1. **Title Slide:** Helio-Track S1 project title, team members, and affiliation.
2. **The Problem:** Data highlighting efficiency loss in fixed systems and user interview findings.
3. **Project Requirements:** Target specifications (accuracy, cost, and power limits).
4. **Concept Selection:** The evaluation matrix showing why we selected the Direct Servo Pillar.
5. **CAD & Tolerancing:** Exploded assembly model and 3D print clearances.
6. **Fabrication:** Materials (PLA/PETG) and physical assembly details.
7. **Testing Results:** Comparative charts of power generation showing a 32% net efficiency boost.
8. **Iterations & Redesign:** Mechanical pivots and software hysteresis algorithms.
9. **Engineering Validation:** Motor torque calculations and PLA stress validation.
10. **Summary & Cost:** Final bill of materials ($43.50 total cost) and project reflections.

2. Mock Presentation Q&A Defense

We anticipated examiner questions and drafted engineering answers:

- **Question:** “Why did you use PETG for the solar panel holder instead of PLA?”

    - **Answer:** “PETG has a higher glass transition temperature (80°C) than PLA (60°C). Since solar panels absorb heat and can get very hot, using PETG prevents the structural holder from softening or warping under hot sunlight.”
- **Question:** “How did you verify the thrust bearing works?”
    - **Answer:** “We measured the current draw of the Azimuth servo. Without the thrust bearing, the servo drew 350mA during slow panning due to friction. With the bearing, current draw dropped to 210mA, confirming less load on the servo.”

3. Calibration Test

We calibrated the system using an LED spotlight. Adjusting LDR bias values in the Arduino code ensured the panel tracked the moving light source smoothly without overshooting or hunting.

DELIVERABLES

- Slide presentation deck outline.
- Mock presentation feedback log.
- Q&A technical defense sheet.

OUTCOME

Students learn how to prepare technical presentations, receive design feedback, and defend engineering decisions under questioning.

WEEK 13

Final Presentation & Demo Day

Design Thinking Phase: Validation

AIM

To showcase the completed dual-axis solar tracking prototype on Demo Day, complete the oral viva voce exam, and submit final reflections.

TOPICS COVERED

- Project Demonstration and Showcase
- Viva Voce Examination
- Examiner Feedback Summary
- Team Reflections

THEORY

Demo Day and the viva voce exam serve as the final validation of the product development process. Students demonstrate their physical prototype, present technical outcomes, and reflect on the project's success.

ACTIVITY

1. Demo Day Showcase Setup

We set up our presentation table with:

- The fully assembled **Helio-Track S1** prototype.
- A printed project poster displaying our CAD drawings, testing graphs, and validation calculations.
- A demonstration rig: We used a high-power LED flashlight to simulate the sun, showing how the solar panel rotated in both axes to lock onto the light beam.

2. Viva Voce Examiner Feedback

- **Examiner Comments:**
  1. The panel holder was praised for its double-shear mounting style, which was noted as a strong improvement over single-cantilever designs.
  2. The examiners suggested that for future commercial scaling, the LDR-based tracking algorithm could be combined with a time-and-latitude astronomical formula to prevent cloud confusion.

- **Result:** The project was approved, with examiners noting the validation calculations in Week 10 as technically complete.

3. Student Reflections

- **Technical Learning:** We gained practical experience in combining Arduino control loop logic with physical 3D-printed parts. Calculating torque and stress proved that simple mechanical decisions can have a huge impact on motor lifespans and structural durability.
- **Collaboration:** Working as an engineering team taught us how to design and build parts simultaneously. Standardizing clearances and screw sizes early prevented assembly errors when putting the final pieces together.
- **HCD Awareness:** Design thinking taught us to start with human needs. Building a solar tracker is not just about writing code; it is about creating an affordable, reliable tool that makes life easier for off-grid users.

DELIVERABLES

- Demo Day showcase setup details.
- Examiner feedback summary.
- Individual and team reflections.

OUTCOME

Students demonstrate full competence in human-centered design, engineering analysis, physical prototyping, and professional defense of their engineering choices.

References & Bibliography

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2. **Osmania University Laboratory Manual (PC453ME):** Academic guidelines for Idea Generation and Product Innovation Laboratory records.
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4. **Arduino Hardware Documentation:** Specifications for the Arduino Nano board, ATmega328P power metrics, and Servo motor pulse-width modulation (PWM) control interfaces.
5. **SolidWorks Design Manual:** DFMA practices and tolerance guidelines for FDM 3D printing assemblies and mechanical joints.
6. **McEvoy, A., Markvart, T., & Castaner, L. (2012). Practical Photovoltaics:** Efficiency calculations and performance characteristics of static vs. tracking solar arrays under varying atmospheric conditions.